Lab Practical 3 (Windows)

To Develop a MapReduce program to implement Matrix Multiplication.

Run this ONLY ONCE when setting up Hadoop for the first time!

Command:

hdfs namenode –format

Starting Hadoop Services (Every Time)

Always open CMD as Administrator before running these commands!

Step 1: Start HDFS

Command:

start-dfs.cmd

Step 2: Start YARN

Command:

start-yarn.cmd

Step 3: Exit Safe Mode

Command:

hdfs dfsadmin -safemode leave

Step 4: Check Running Services

jps

Practical 3

Step 1: Create Working Directory

Commands:

```
mkdir C:\matrixmultiply
cd C:\matrixmultiply
```

Step 2: Create Combined Input File (input.txt)

Command:

```
notepad input.txt
```

Enter the following content (format: matrix_name,row,column,value):

```
M,0,0,1
M,0,1,2
M,0,2,3
M,1,0,4
M,1,1,5
M,1,2,6
N,0,0,7
N,0,1,8
N,1,0,9
N,1,1,10
N,2,0,11
N,2,1,12
```

Step 3: Create Multiply.java

Command:

```
notepad Multiply.java
```

Paste this code:

```
import java.io.IOException;
import java.util.*;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MatrixMultiply {

  public static class Map extends Mapper<LongWritable, Text, Text, Text> {
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

      String[] val = value.toString().split(",");

      if (val[0].equals("A")) {
        for (int k = 0; k < 2; k++) {
          context.write(new Text(val[1] + "," + k),
              new Text("A," + val[2] + "," + val[3]));
        }
      }

      if (val[0].equals("B")) {
        for (int i = 0; i < 2; i++) {
          context.write(new Text(i + "," + val[2]),
              new Text("B," + val[1] + "," + val[3]));
        }
      }
    }
  }

  public static class Reduce extends Reducer<Text, Text, Text, IntWritable> {
    public void reduce(Text key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

      int[] a = new int[2];
      int[] b = new int[2];

      for (Text val : values) {
        String[] data = val.toString().split(",");

        if (data[0].equals("A"))
          a[Integer.parseInt(data[1])] = Integer.parseInt(data[2]);

        else
          b[Integer.parseInt(data[1])] = Integer.parseInt(data[2]);
      }

      int sum = 0;

      for (int i = 0; i < 2; i++)
```

```
        sum += a[i] * b[i];

      context.write(key, new IntWritable(sum));
    }
  }

  public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "Matrix Multiplication");

    job.setJarByClass(MatrixMultiply.class);
    job.setMapperClass(Map.class);
    job.setReducerClass(Reduce.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

Step 4: Compile the Java Program
Command:

```
javac -classpath "%HADOOP_HOME%\share\hadoop\common\hadoop-common-
3.3.6.jar;%HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-client-core-
3.3.6.jar;%HADOOP_HOME%\share\hadoop\common\lib\commons-cli-1.2.jar" MatrixMultiply.java
```

Step 5: Create JAR File
Command:

```
jar cf multiply.jar Multiply*.class
```

Step 6: Upload Input File to HDFS
Commands:

```
hadoop fs -put C:\matrixmultiply\input.txt /
hadoop fs -ls /
```

Step 7: Delete Old Output (if exists)
Command:

```
hadoop fs -rm -r /matrix_output
```

Step 8: Run the MapReduce Job
Command:

```
hadoop jar multiply.jar Multiply /input.txt /matrix_output
```

Step 9: View the Output
Command:

```
hadoop fs -cat /matrix_output/part-r-00000
```

Stopping Hadoop Services

Always stop Hadoop properly before closing CMD windows!

Step 1: Stop YARN

```
stop-yarn.cmd
```

Step 2: Stop HDFS

```
stop-dfs.cmd
```